Oscillations

SYNOPSIS

Repetitive/cyclical disturbances propagating through space and time are waves. Some waves are familiar: Ocean waves, waves on a string, and sound waves are disturbances that travel from a place and time of creation to an observer. Mathematically, cosinusoidal functions encapsulate such disturbances, i.e., some combinations of sines and cosines. For example, we can write the amplitude of ocean waves as a function of space and time:

$$A(x, t) = A_0 \sin(kx - \omega t) \tag{1}$$

where A is the amplitude, A_0 is the maximum amplitude, k is the wave number (1/m), and ω is the angular frequency (rad/s). This is a moving wave traveling along the x-axis. The wave number, k, and angular frequency, ω, are parameters fixed by the wavelength and period (thus frequency) of the moving wave, respectively. This is a general expression for a moving wave, and we will delve into this equation further shortly.

BEFORE YOU READ ON — PREDICT

Hang a mass on a spring, let it settle, then pull it down a little further and release it. Gravity is pulling on that mass the whole time. So when you write down the equation that governs the oscillation, **does *g* appear in it?** Commit to yes or no in your guided notes. The answer is the whole point of the derivation below, and most people get it wrong the first time.

PERFORMANCE EXPECTATIONS

1. **State** Hooke's law for a spring and identify the restoring force in a mass-spring system.
2. **Combine** Hooke's law with Newton's second law to reach $ma = -kx$, and say what the minus sign does to the motion.
3. **Recognise** $x(t) = A\cos(\omega t + \varphi)$ as the motion this produces, and identify A, ω and φ on a graph.
4. **Explain** why a force proportional to displacement and opposite in sign produces repetition rather than a one-way motion.
5. **Decide** whether g appears in the equation of motion for a hanging mass, and justify the answer.

Instructions and Questions

For now, we examine a simpler system: Spring-mass oscillation. When you hang a mass on a spring, stretch, and let go, the spring-mass system oscillates back and forth. If you measured the amplitude of this disturbance as a function of

time, i.e., stretching and compression of the spring as a function of time, you see that the following function can describe such disturbance:

$$A(t) = A_0 \sin(\omega t) \tag{2}$$

Here, you see that the sine is only a function of time, and the kx term is not in the sine function argument. This is simply because the spring-mass system is not moving, i.e., the setup is fixed in space. So, you can see that this is a simpler system.

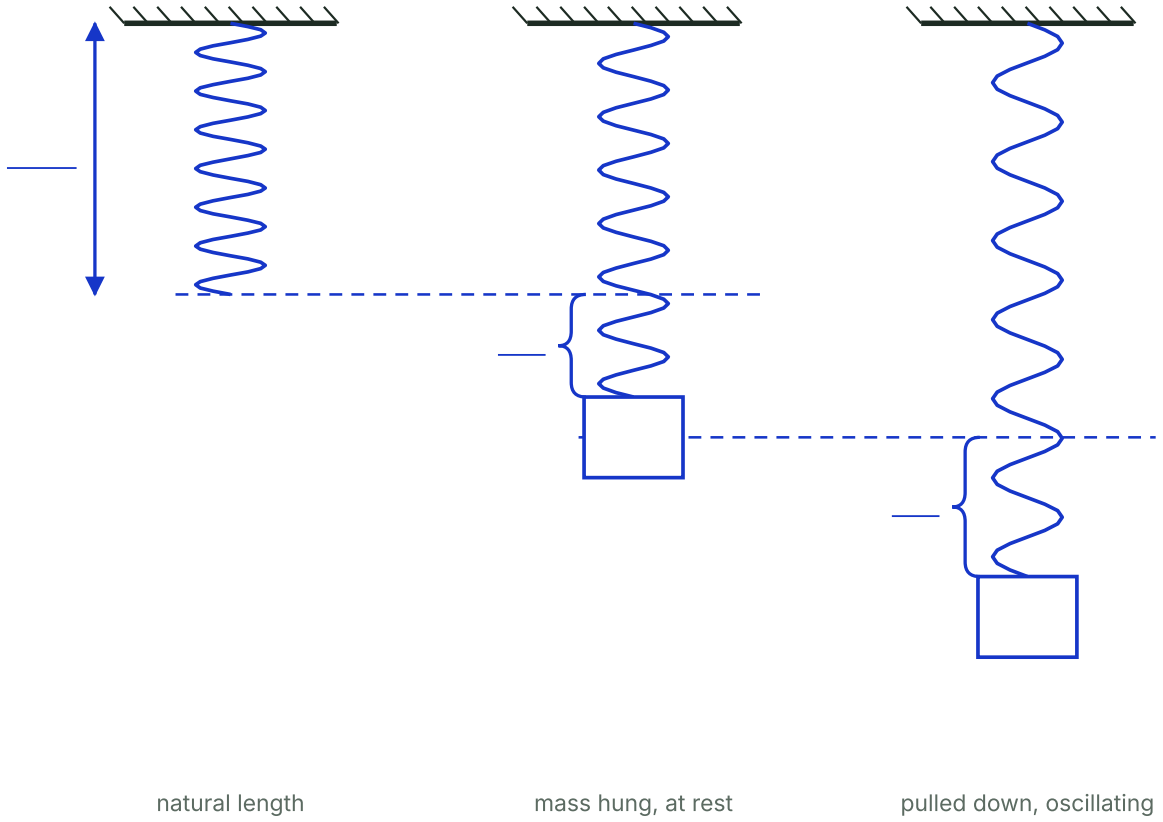
The spring-mass oscillation is known as **simple harmonic motion (SHM)**. SHMs are ubiquitous. Let's look at the physics that underpins SHM. The idea is simple: When a spring is stretched or compressed, a "restoring" force tries to pull it back to its natural length. The further the spring is stretched/compressed, the larger the restoring force. In other words, the larger the displacement, the larger the restoring force. Both the displacement and force are vector quantities. Notice that this restoring force, therefore, always points in the opposite direction to the displacement. **Hooke's law** describes the relationship between the displacement and the restoring force:

$$F_\mathrm{k} = -k_\mathrm{s} x' \tag{3}$$

Where F_k is the restoring force (N, the subscript k represents that the force has to do with a spring), k_s is the spring constant, and x' is the displacement (see the figure below). Don't confuse this k_s, the spring constant, with k, the wavenumber.

Imagine a mass hanging at one end of a spring. The spring will elongate and settle at a length. Here, $F_\mathrm{k} = F_\mathrm{g}$, where F_g is the weight of the mass. Let this elongation be x_0, so $x_0 = L - L_0$, where L_0 is the natural length, and L is the length when mass is hung. At this point, the mass is not oscillating.

Now, if we pull the mass further down (or push it up) and let go, the mass oscillates. This oscillation is a simple harmonic motion. We will analyze the forces below and derive Eq. (2) by solving the equation of motion.



The three states the derivation needs. **Down is positive.** x_0 is how far the mass stretches the spring just by hanging there; x is how much further you pull it; and the spring itself only ever knows the total, $x' = x_0 + x$.

First, the mass hanging at rest

1. Write Newton's second law for the middle picture, where the mass hangs without oscillating. What is F_{net}, and what does that tell you about F_{k} and F_{g}?
2. Put Hooke's law and the weight into that statement and solve for the stretch x_0.

Now pull it down and let go

3. Write Newton's second law for the right-hand picture. The spring is now stretched by the *total* $x' = x_0 + x$, so what is the new spring force F'_k? What is F_g?

4. Substitute both into $ma = \Sigma F$ and expand the bracket. Two of the three terms cancel. Which two, and what lets you cancel them?

THE PUNCHLINE

The acceleration — and so the force — is proportional to the displacement, and points the opposite way. That one sentence is the definition of simple harmonic motion. Any system you can get into this form will oscillate, and it will oscillate as a sine. That is why Eq. (2) looks the way it does, and it is why Note 5 can run the same argument on a pendulum and get the same answer.

WITH CALCULUS

Writing it as an equation of motion

$ma = -kx$ says the force is proportional to the displacement. To get from there to the sine of Eq. (2) rather than merely asserting it, write the acceleration as what it actually is — the second derivative of position. One subtlety: the position of the mass is x', not x.

5. Start from $a = \mathrm{d}^2x'/\mathrm{d}t^2$ with $x' = x_0 + x$, and show that it equals $\mathrm{d}^2x/\mathrm{d}t^2$. Say which property of x_0 you used.

6. Write the result of step 4 as a differential equation, and say what kind of function could satisfy it.

Check yourself

7. A mass hangs at rest on a spring, stretching it by 4.0 cm. You pull it down a further 3.0 cm. By how much is the *spring* stretched at that moment, and what is the displacement that appears in $ma = -kx$?

8. Two students derive the equation of motion. One measures displacement from the spring's natural length, the other from the hanging rest position. One of them ends up with a g in the answer. Which one, and is that student wrong?

9. In one sentence: why does a stiffer spring make the oscillation faster? Argue it from $ma = -kx$ alone.

Looking ahead

You have the shape of simple harmonic motion and the one sentence that defines it. What you do not yet have is **how fast** it oscillates — nothing above gives a period or a frequency. **Note 5** gets it, by solving the equation of motion rather than describing it, and then runs the same argument twice more: on a pendulum, where the restoring force comes from gravity rather than a spring, and on a damped oscillator, where the amplitude dies away.

Where this sits. This is Module Note 7's $F = ma$ applied to one particular force law, and nothing more. Everything characteristic of oscillation comes from the minus sign in Hooke's law — that the force points back toward where you came from.

BACK TO YOUR PREDICTION

No, g does not appear. It is in the derivation right up until the second-to-last line and then cancels, against the stretch it caused in the first place. Hanging the mass moved the rest position down by x_0; it did not change how the system oscillates about that position. A spring-mass oscillator on the Moon, hung from the same spring, has the same period.